Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software Engineer with 2+ years turning **business requirements** into tested software, combining SQL data conversions, **unit and integration testing**, **production monitoring**, and **root-cause debugging** across Azure and AWS systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Web Technologies, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *LA, California*

- Optimized production query patterns and indexing for a multi-tenant platform on **Azure SQL** and Blob Storage serving 500+ users, isolating performance bottlenecks to cut dashboard latency from 1.2s to 350ms.
- Built **data conversion** and integration pipelines across PostgreSQL, SQL Server, and cloud storage with validation and reconciliation checks, processing 5K+ monthly jobs through CI/CD releases at 99%+ production reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *LA, California*

- Translated instructor workflows into React and Flask features for question upload, rubric generation, and grading history, serving 150+ instructors and 10K+ daily requests while cutting feedback turnaround 60%.
- Resolved OCR reliability and timeout issues in concurrent batch grading with Claude Vision fallback paths, ThreadPoolExecutor processing, secure file ingestion, and per-call **cost logging** across 1K+ files per batch.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Contributed from **requirements analysis** through deployment on cross-platform data-sharing microservices, converting raw JSON, CSV, and Parquet data with AWS Glue and S3, improving response times 30% and pipeline efficiency 68%.
- Automated nightly **Jenkins** jobs that built executables, installed them on test VMs, and ran **unit and integration tests** across Windows, Linux, and macOS, generating CSV reports and failure logs for triage.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Built a **resource management platform** with MERN, REST APIs, and WebSockets for resource allocation, customer records, billing, and lead workflows, synchronizing 2M+ records across locations and reducing resource wastage 42%.
- Monitored production with **Sentry**, Azure Monitor, and Kibana while gating Jenkins releases on unit, integration, and end-to-end tests, achieving 95%+ successful build validation across Docker-based staging deployments.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Translated client **business requirements** into an end-to-end system design, mapped 12+ backend integration points, and presented the architecture to the client for alignment, reducing design errors 25% before implementation.
- Developed Flask APIs for booking, pricing, coupons, and payment webhooks with Firebase real-time sync, sustaining sub-200ms responses across 50K+ records and reducing manual refund handling by about 70%.

Skills

Analysis & Testing: Requirements Analysis, System Design, Technical Documentation, Stakeholder Communication, Agile/Scrum, SDLC, Unit Testing, Integration Testing, End-to-End Testing, PyTest, Mockito, Jest, Selenium, Debugging, Root-Cause Analysis, Sentry, Azure Monitor, ETL Pipelines
Languages & Data: Python, Java, JavaScript, TypeScript, C++, SQL, PostgreSQL, SQL Server, Azure SQL, MySQL, MongoDB
Cloud & Tools: Azure, AWS (S3, RDS, Glue), Docker, Jenkins, GitHub Actions, CI/CD, Git, Linux, REST APIs, Flask, React.js

Projects

Chat2SQL Code | *Java, TypeScript, SQL, PostgreSQL, MySQL, Docker, Maven*

- Extended an AI-assisted **SQL** workspace in Java and TypeScript supporting 10+ database systems, enabling schema synchronization, query authoring, and data management from one cross-platform interface deployed with Docker.

HMIS - Hospital Management Information System Live | *TypeScript, MERN, ABDM APIs, Twilio, Sentry*

- Built a hospital operations platform with scheduling dashboards, consent-based ABDM record access, Twilio reminders, and **Sentry monitoring**, supporting 100+ daily appointments and reducing administrative workload 30% for staff.

CPU-scheduling Simulator Live | *C++, React.js, JavaScript, EJ-2*

- Implemented and **tested** FCFS, SJF, SRT, Round Robin, priority, and MLQ scheduling across preemptive and non-preemptive scenarios with Gantt visualization, reducing average turnaround time 25% and debugging time 20%.